

Assignment 1

Due: February 22, 2026 at 11:59 pm (Central Time)

Q1. Please write python codes using the NumPy library (when applicable) to answer the following questions and submit the python source codes on Canvas.

- (a) (5%) Write a python code to create a NumPy array, called `a_col`, of shape (5,1) from the following list: [5, 9, 2, 4, 8, 9, 6, 1, 3, 11]
- (b) (5%) Write a python code to sort `a_col` in the descending order.
- (c) (10%) Write a python code to compute the z-score normalized version of `a_col` using the following formula:

$$a_{norm} = \frac{a_col - \text{mean}(a_col)}{\text{std}(a_col)}$$

- (d) (10%) Write a python code to create a NumPy array, called `a_clipped`, from `a_col` where:
 - Values below the median are replaced by the median
 - Values greater than or equal to the median remain unchanged
- (e) (10%) Write a python code to extract the subarray from index 2 to 6 (inclusive) from `a_col`, reverse it, and then select every other element from the reversed subarray, and print the result.

Q2. Consider the following two sample tables: DeptTable and EmployeeTable. DeptTable consists of all departments' ID, name, and start year. EmployeeTable consists of all employees' ID, department ID, highest educational degree, evaluation, and age. Please write python codes (using Pandas and/or NumPy libraries) for parts (a) to (g) and use table operations for parts (b) to (g) when applicable. Please submit the python source codes on Canvas.

Table 1. DeptTable

Dept_ID	Dept_Name	Dept_StartYear
1003	A	2011
1001	B	2003
1002	C	2007

Table 2. EmployeeTable

Emp_ID	Dept_ID	Degree	Eval	Age
Emp1	1001	Master	3.7	33
Emp2	1002	Master	3	25
Emp3	1002	Doctoral	2.2	38
Emp5	1001	Master	2.7	42
Emp6	1002	Bachelor	1.0	31

- (a) (5%) Write a python code to create a separate Pandas DataFrame for each of the above tables: i.e., DeptTable and EmployeeTable.
- (b) (5%) Write a python code that returns the name of the newest department.

- (c) (10%) Write a python code that returns the name of the departments that have more than two employees.
- (d) (10%) Write a python code that returns the name of the department that has the smallest number of employees along with its number of employees.
- (e) (10%) Write a python code that returns the average age of employees with the Master's degree for each department. The expected output is a table (pandas DataFrame) with two columns: Dept_ID and AvgAge. If a department does not have any employee with a Master's degree, the AvgAge for that department should be "NA" in the output table.
- (f) (10%) Write a python code that creates a new column, called LetterEval, in EmployeeTable by applying the following mapping on the Eval column:
- $\text{Eval} > 3.5 \rightarrow \text{A}$
 - $2 < \text{Eval} \leq 3.5 \rightarrow \text{B}$
 - $\text{Eval} \leq 2 \rightarrow \text{C}$
- (g) (10%) Write a python code that returns the number of employees at each of the following two age ranges, i.e., [30, 40) and [40, 50), for each department. The expected output is a table (Pandas DataFrame) with three columns: Dept_ID, AgeRange, and NumberOfEmployees.